

JEMBY DIGITAL TWIN: Panel Sizing, Phase Rules, & DOT Layout Specification

This specification defines the deterministic sizing, phase allocation, multi-pole cell spanning, and Graphviz DOT table generation rules for commercial, industrial, and residential electrical panelboards.

1. Core Sizing Rules: Multiples of 6 & 12

Commercial and industrial panelboards (Square D NQ/NF, Siemens P1, Eaton Pow-R-Line, ABB/GE Pro-Stock) are **3-phase or split-phase panels arranged in 2 columns** (Odd circuits left, Even circuits right).

1.1 Standard Space Counts (N)

- Standard Space Counts:

$N \in \{18, 30, 42, 54, 72, 84\}$ (Split-phase residential/light commercial may also use $N \in \{12, 24, 32, 40\}$).

- Total Row Count:

$Rows = (N / 2)$

Every 3-phase commercial panel has a row count divisible by 3 (or an odd/even multiple) because it cycles continuously across the 3 phase pairs: **Row Phase Pairs:** (A, A), (B, B), (C, C), (A, A), (B, B), (C, C), ...

1.2 Digital Twin Auto-Sizing Algorithm

Given a set of required branch loads requiring P_{used} total breaker poles: 1. **Target Sizing with Growth Buffer:** Calculate target spaces with a minimum 20% growth buffer: $N_{target} \geq P_{used} \times 1.20$ 2. **Standard Frame Selection:** Select the smallest standard enclosure space count: $N = \min \{s \in \{18, 30, 42, 54, 72, 84\} \mid s \geq N_{target}\}$ 3. **Spare Auto-Fill:** Any unassigned slots from $P_{used} + 1 \dots N$ are automatically instantiated as [SPARE] entries.

2. Phase Staggering & Multi-Pole Ganging Rules

2.1 3-Phase Panel Bus Stab Alternation (208Y/120V & 480Y/277V)

LEFT (ODD) RIGHT (EVEN)					
■ Slot 1 (A) ■ ■ ■ (A) ■ ■ ■ Slot 3 (B) ■ ■ ■ (B) ■ ■ ■ Slot 5 (C) ■ ■ ■ (C) ■ ■ ■ Slot 7 (A) ■ ■ ■ (A) ■ ■ ■ Slot 2 (A) ■ (Row 1) ■ Slot 4 (B) ■ (Row 2) ■ Slot 6 (C) ■ (Row 3) ■ Slot 8 (A) ■ (Row 4)					
Row Number	Left Pole (Odd)	Right Pole (Even)	Bus Phase	Phase Voltage (V _{L-N})	Line-to-Line (V _{L-L})
Row 1	Slot 1	Slot 2	Phase A	120V / 277V	208V / 480V
Row 2	Slot 3	Slot 4	Phase B	120V / 277V	208V / 480V
Row 3	Slot 5	Slot 6	Phase C	120V / 277V	208V / 480V

Row 4	Slot 7	Slot 8	Phase A	120V / 277V	208V / 480V
Row 5	Slot 9	Slot 10	Phase B	120V / 277V	208V / 480V
Row 6	Slot 11	Slot 12	Phase C	120V / 277V	208V / 480V

2.2 Split-Phase Panel Bus Stab Alternation (120/240V 1-Phase 3-Wire)

For single-phase systems, rows alternate strictly between Phase A and Phase B: **Row r \implies Phase A if r is odd, else Phase B**

2.3 Multi-Pole Breaker Ganging & Cell Spanning Rules

- **1-Pole Breaker (120V / 277V)**: Occupies 1 slot cell (e.g., Slot 1). Connected to 1 phase stab.
- **2-Pole Breaker (208V / 240V / 480V)**: Occupies 2 vertically adjacent slots on the same side (e.g., Slots 13 & 15 on Left, or 14 & 16 on Right).
- **Table Render**: The load description cell uses ROWSPAN="2" starting at the top slot.
- **Numeric Slots**: Both numeric slot IDs (e.g. 13 and 15) render in their respective rows.
- **Phase Spanning**: Spans Phase A + B, Phase B + C, or Phase C + A.
- **3-Pole Breaker (208V 3Ø / 480V 3Ø)**: Occupies 3 vertically adjacent slots on the same side (e.g., Slots 1, 3, 5 or 2, 4, 6).
- **Table Render**: The load description cell uses ROWSPAN="3" starting at the first slot.
- **Numeric Slots**: All 3 numeric slot IDs render in their respective rows.
- **Phase Spanning**: Spans Phase A + B + C across all three bus stabs.

3. Graphviz HTML Table Specification

3.1 Table Schema Architecture

The central panel node is rendered as an HTML-like table record with 4 principal columns: 1. **Column 1 (LEFT_LOAD)**: Left Breaker status, circuit name, and rating (with ROWSPAN for multi-pole). 2. **Column 2 (LEFT_NUM)**: Left Slot Number (Odd: 1, 3, 5, \dots). 3. **Column 3 (RIGHT_NUM)**: Right Slot Number (Even: 2, 4, 6, \dots). 4. **Column 4 (RIGHT_LOAD)**: Right Breaker status, circuit name, and rating (with ROWSPAN for multi-pole).

3.2 Visual Styling Tokens

- **Panel Enclosure Border**: COLOR="#1A365D", BORDER="2", CELLBORDER="1", CELLSPACING="0".
- **Primary Header (Panel Title)**: BGCOLOR="#1A365D", .
- **Secondary Header (Mains Rating)**: BGCOLOR="#2A4365", .
- **Numeric Slot Columns**: BGCOLOR="#E2E8F0", 1, 2.
- **Active Breakers ([ON])**: Standard background (#FFFFFF), bold slot tag, trip rating in parentheses.
- **Spare Slots ([SPARE])**: Highlight background BGCOLOR="#FEFCBF", [SPARE] Unused :N.
- **Spaces ([SPACE])**: Highlight background BGCOLOR="#F1F5F9", [SPACE] Blank.

3.3 Reference DOT Syntax

```
digraph PanelTwin {
  graph [rankdir=TB, splines=polyline, nodesep=0.8, ranksep=0.8, fontname="Helvetica"];
  node [shape=none, fontname="Helvetica"];
  edge [penwidth=1.2, color="#1E293B", fontname="Helvetica", fontsize=8];

  Panel_MDP [label=<
    <TABLE BORDER="2" CELLBORDER="1" CELLSPACING="0" COLOR="#1A365D" BGCOLOR="#FFFFFF">
      <!-- Header: Panel Title -->
      <TR>
        <TD COLSPAN="4" BGCOLOR="#1A365D" PORT="main">
          <FONT COLOR="WHITE" POINT-SIZE="11"><B>MAIN DISTRIBUTION PANEL (42 SPACES)</B></FONT>
        </TD>
      </TR>
      <!-- Subheader: Mains & Voltage -->
      <TR>
        <TD COLSPAN="4" BGCOLOR="#2A4365">
          <FONT COLOR="WHITE" POINT-SIZE="9"><B>[ 400A MAIN BREAKER - 120/208V 3Ø 4W ]</B></FONT>
        </TD>
      </TR>

      <!-- Row 1: (1P Left, 1P Right) -->
      <TR>
        <TD ALIGN="LEFT" PORT="ckt_1">[ON] 1: Lighting Cir 1 (20A)</TD>
        <TD BGCOLOR="#E2E8F0"><B>1</B></TD>
        <TD BGCOLOR="#E2E8F0"><B>2</B></TD>
        <TD ALIGN="RIGHT" PORT="ckt_2">(20A) Receptacles Lab :2 [ON]</TD>
      </TR>

      <!-- Row 2-4: (3P Left spanning 3,5,7; Right has 1P, SPARE, SPARE) -->
      <TR>
        <TD ALIGN="LEFT" ROWSPAN="3" PORT="ckt_3_5_7"><B>[ON] 3,5,7: RTU-1 (3P 50A)</B></TD>
        <TD BGCOLOR="#E2E8F0"><B>3</B></TD>
        <TD BGCOLOR="#E2E8F0"><B>4</B></TD>
        <TD ALIGN="RIGHT" PORT="ckt_4">(20A) Server Room Dedicated :4 [ON]</TD>
      </TR>
      <TR>
        <TD BGCOLOR="#E2E8F0"><B>5</B></TD>
        <TD BGCOLOR="#E2E8F0"><B>6</B></TD>
        <TD ALIGN="RIGHT" BGCOLOR="#FEFCBF" PORT="ckt_6"><FONT COLOR="#744210">[SPARE] Unused :6</FONT></TD>
      </TR>
      <TR>
        <TD BGCOLOR="#E2E8F0"><B>7</B></TD>
        <TD BGCOLOR="#E2E8F0"><B>8</B></TD>
        <TD ALIGN="RIGHT" BGCOLOR="#FEFCBF" PORT="ckt_8"><FONT COLOR="#744210">[SPARE] Unused :8</FONT></TD>
      </TR>
    </TABLE>
  >];
}
```

4. Generator Prompt Directive

> "For commercial and industrial panel twins, constrain total circuit spaces N strictly to \{18, 30, 42, 54, 72, 84\}. Model the interior schedule as an N/2-row two-column array (odd circuits left, even circuits right). Group multi-pole loads using vertical cell spanning (ROWSPAN=2 or ROWSPAN=3 on the load column) while always rendering all numeric pole IDs (1 to N). Render unoccupied slots with a yellow highlight (#FEFCBF) marked [SPARE]."